INCH-POUND

MIL-DTL-38359E
27 September 2011
SUPERSEDING
MIL-DTL-38359D
24 January 2011

DETAIL SPECIFICATION

CABLE, POWER, ELECTRICAL, AIRPORT LIGHTING, CROSS-LINKED, POLYETHYLENE (XLP)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 8 AWG single conductor electric power cable of two voltage ranges intended primarily for use in airport lighting series circuits.

1.2 Classification. Cables are of the following classes, as specified:

Class 1 – 3,000 V rms
Class 2 – 5,000 V rms

1.2.1 Part or Identifying Number (PIN). PINs are of the following form, as specified:

Class 1 – M38359-3K
Class 2 – M38359-5K

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAI, P.O. Box 3990, Columbus, Ohio 43218-3990, or email to WireCable@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the documents are the issues of the documents cited in the solicitation or contract.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 10012-1	-	Equipment, Quality Assurance Requirements for Measuring - Part 1 Metrological Confirmation System for Measuring Equipment

(Copies of these documents are available online at http://www.iso.ch or from the International Organization for Standardization American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036.)

NCSL INTERNATIONAL

NCSL Z540.3	-	General Requirements for Calibration of Measuring and Test Equipment

(Copies of these documents are available online at http://www.ncsli.org or from NCSL International, 2995 Wilderness Place, Suite 107 Boulder, Colorado 80301-5404)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION, (NEMA)

NEMA WC 71	–	Standard for Non-Shielded Power Cables Rated 2001 – 5000 Volts for Use in the Distribution of Electrical Energy

Copies of these documents are available online at http://www.nema.org or from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.)

2.4 Order of precedence. In event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. Cables specified under this specification shall be subjected to a first article inspection (see 6.6). First article inspection shall be performed at laboratory acceptable to the procuring activity on sample units produced with equipment and procedures used in production.

3.2 Materials. The materials for the principal components of the cable shall be as specified herein. Prior approval to use substitute material must be obtained from the qualifying activity. When a definite material is not specified, a material shall be used that will enable the finished products to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product. All materials used in the manufacture of cables furnished to comply with this specification shall be of such quality and form that the finished product conforms to the requirements of this specification.

3.3 Design and construction. The cable shall be designed and constructed for use in airport lighting systems. It shall be constructed for continuous outdoor service under all weather conditions while on the surface of the ground, buried directly in the ground, installed in conduit, or submerged in fresh or salt water. The voltage rating, high potential withstanding voltage, wire size, and average thickness of the insulation for the cables specified herein shall be as specified in table I.

TABLE I. Combined cable characteristics.

Characteristic	Class 1	Class 2
Voltage rating	3,000 V rms	5,000 V rms
High potential withstanding voltage	7,500 V rms (5 minutes)	11,000 V rms (5 minutes)
Insulation thickness (average)	.075 inch (1.91 mm)	.11 inch (2.79 mm)
Diameter:		
Minimum	.31 inch (7.87 mm)	.38 inch (9.65 mm)
Nominal	.32 inch (8.13 mm)	.39 inch (9.91 mm)
Maximum	.34 inch (8.64 mm)	.41 inch (10.41 mm)

3.3.1 Conductors. The conductor shall conform to type C, number 8 AWG, class B, tin coated.

3.3.1.1 Splices. Splices in the conductor shall be such that each splice will not increase the diameter of the conductor. No splice shall be made in the conductor as a whole. Not more than one strand shall be spliced in any one foot of conductor.

3.3.1.2 Resistance. The resistance of the conductor of 1,000 feet of completed cable at +20 ºC shall not be greater than .65 ohm.

3.3.1.3 Conductor shield. A semiconducting layer or tape shall be applied directly over the stranded conductor, over which the cable insulation shall be applied. The conductor shield shall have a minimum thickness of .005 inch (0.13 mm).

3.3.1.4 Marker threads. Nonhygroscopic-type marker threads shall be provided in the conductor strands or directly along the conductor for identification purposes. The color and number of threads used shall be identical with those used by the manufacturer on his other products.

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3.3.2 Insulation. The insulation shall consist of a black cross-linked thermoplastic polyethylene (XLP) compound which meets the requirements for the insulation specified in NEMA WC 71 and all the requirements specified herein. The insulation shall be applied tightly and concentrically about the conductor over the conductor shield and shall be free from pores, splinters, and other such defects.

3.3.2.1 Insulation thickness. The average thickness of the insulation shall be not less than that specified in table I. The minimum thickness at any point shall be not less than 90 percent of the thickness specified in table I. If the insulation is applied in more than one layer, the adjacent layers shall be molded into a homogeneous mass and the mass considered as a whole for all measurements and tests.

3.3.2.2 Insulation resistance. The measured insulation resistance (see 4.6.4) at 15.6 ºC shall be not less than that calculated from the following formula:

$$R = K \log_{10} \frac{D}{d}$$

Where R = Megohms - 1000 feet

D = Outside diameter of insulation in inches

d = Diameter of uninsulated conductor in inches

K = Constant 25,000 for insulation specified herein

NOTE: If the temperature differs from 15.6 ºC during the resistance measurement, correction shall be made by multiplying the calculated value of R by the coefficient corresponding to the temperatures during measurement. The coefficient for the insulation shall be that specified by the manufacturer. The temperature coefficient shall be determined in accordance with NEMA WC 71.

3.3.2.3 Joints. All joints made in the insulation shall be such that all parts affected in the process will be electrically and physically equal to the remainder of the insulation and shall conform to the thickness requirements specified herein.

3.3.2.4 Physical properties. The physical properties of the insulation shall be as follows:

a. Original physical requirements:

(1) Tensile strength, initial, 1,800 lbf/in^2 minimum.

(2) Elongation at rupture, 350 percent minimum.

(3) Modulus at 200 percent elongation, 1,000 lbf/in^2 minimum.

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b. Aging requirements:

(1) After air oven test at a minimum of 150 ºC for 168 hours:

(a) Tensile strength in percent of unaged value, 75 percent minimum.

(b) Elongation at rupture in percent of unaged value, 65 percent minimum.

(2) After air pressure heat test for 42 hours at 80 lbs/in^2 and 127 ºC:

(a) Tensile strength in percent of unaged value, 75 percent minimum.

(b) Elongation at rupture in percent of unaged value, 75 percent minimum.

3.3.2.5 Ozone resistance. The cable insulation shall be resistant to ozone attack.

3.3.2.6 Moisture absorption. The moisture absorption properties of the cable insulation shall be as follows:

Increase in capacitance:

1 to 14 days: 3 percent maximum

7 to 14 days: 1.5 percent maximum

3.3.2.7 Cold bend. The cable insulation shall not crack as a result of bending at a radius of 2.5 times the cable diameter when in an ambient temperature of -40 ºC or less.

3.4 Performance.

3.4.1 High temperature. The cable shall be resistant to high temperatures encountered while in storage and under service conditions. When in an ambient temperature of 71 ºC, a full reel of cable shall not suffer permanent set to any portion of the cable or show evidence of insulation deterioration.

3.5 Surface marking on cable. The manufacturer's name and contract number shall be legibly marked on the surface of all cable. The markings shall be of paint or ink applied at intervals not exceeding 10 feet. The markings shall be durable and not readily smeared during handling of the cable.

3.6 Workmanship. Workmanship shall be of the quality necessary to produce cable free from all defects which affect proper functioning in service. The exterior surface of the cable shall be smooth, uniform, and free from splinters, ridges, grooves, indentations, and protuberances visible to the naked eye.

3.7 Cable lengths. Except as specified in, cable shall be of one continuous 3,000-foot length, +10, -3 percent (see 6.2).

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3.7.1 Allowable short lengths. Unless otherwise specified by the acquiring activity (see 6.2), short lengths may be supplied in accordance with the following schedule on orders of 100,000 feet and over:

a. 85 percent or more of the order: 3,000-foot lengths.

b. 10 percent or less of the order: 1,500-foot minimum lengths.

c. 5 percent or less of the order: 1,000-foot minimum lengths.

For orders of less than 100,000 feet, all lengths shall be 3,000 feet +10, -3 percent.

3.8 Recycled, recovered or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.9 Pure tin. The use of pure tin is prohibited (see 6.7). The maximum amount of pure tin is 97 percent and the alloy material shall inhibit the growth of tin whiskers.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted at a temperature of 25±10 ºC.

4.3 First article inspection. The first article inspection shall consist of all the tests specified in 4.5.

4.3.1 Sample size. The sample shall consist of a reel of cable or a 50 foot or longer coil. The sample shall be appropriately identified with the manufacturer's part number and other information as required by the acquiring activity.

4.3.2 Requirements cross-reference matrix. Table II provides a cross-reference matrix of section 3 requirements tested or verified below.

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TABLE II. Requirements cross-reference matrix.

Requirement	Verification	Requirement	Verification
3.1	4.3	3.3.2.3	4.6.2, 4.6.4 , 4.6.4.1
3.3	4.6.2, 4.6.3	3.3.2.4	4.6.10
3.3.1	4.6.2	3.3.2.5	4.6.7
3.3.1.1	4.6.2	3.3.2.6	4.6.6
3.3.1.2	4.6.9	3.3.2.7	4.6.8
3.3.1.3	4.6.2	3.4.1	4.6.11
3.3.1.4	4.6.2	3.5	4.6.2
3.3.2	4.6.2	3.6	4.6.2
3.3.2.1	4.6.4.1	3.7	4.6.2
3.3.2.2	4.6.4	3.7.1	4.6.2

4.4 Conformance inspection. Conformance inspection shall consist of group A and B inspections.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B.

4.4.1.1 Selection of sample units. The inspection sample shall be product selected at random from the lot without regard to quality and shall be of the size specified in table V. A sample unit is defined as a length of cable drawn from a unit of product.

4.4.1.2 Test specimen. A test specimen may be the entire sample unit (length of finished cable) or any portion of the sample unit that is to be tested.

4.4.2 Group A inspection. Each cable shall be subjected to the individual tests shown in table III.

TABLE III. Group A inspection.

Inspection	Requirement	Test method
Visual inspection	3.3	4.6.2
Inspection of product	3.6	4.4.1, 4.4.1.1, 4.5.1
High potential withstanding voltage test	3.3	4.6.3
Insulation resistance test	3.3.2.2	4.6.4
Continuity test	3.3	4.6.5

4.4.3 Group A acceptance. Each sample selected in accordance with 4.4.1.1 shall be examined to determine conformance with the requirements in table III. If one or more defects are found in the inspection sample, the production lot shall be rejected and not supplied to this specification.

4.4.4 Group B inspection. Group B inspection shall consist of the tests specified in table IV, in the order shown. Group B inspection shall be made on sample units selected from inspection lots which have passed group A inspection. Sample size selection shall be in accordance with table VI (small sample inspection).

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TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Method paragraph
Moisture absorption	3.3.2.6	4.6.6
Ozone resistance test	3.3.2.5	4.6.7
Cold bend test	3.3.2.7	4.6.8
Insulation resistance test	3.3.2.2	4.6.4
Insulation thickness test	3.3.2.1	4.6.4.1
Wire resistance test	3.3.1.2	4.6.9
Physical properties test	3.3.2.4	4.6.10
High temperature test	3.4.1	4.6.11

4.4.4.1.1 Inspection lot. An inspection lot shall consist of all cables of the same type designation (see 1.2.1) produced under essentially the same conditions and offered for inspection at the same time.

TABLE V. Inspection sample.

Production lot size[1/]	Sample size
2 to 8	2
9 to 15	3
16 to 25	5
26 to 50	8
51 to 90	13
91 to 150	20
151 to 280	32
281 to 500	50
501 to 1200	80
1201 to 3200	125
3201 to 10000	200
1001 to 35000	315

1/ Lot size will be based on number of reels, spools or coils of product.

TABLE VI. Small sample inspection.

Lot size (passed group A inspection)	Sample size
2 to 25	2
26 to 150	3
151 to 1200	5

4.4.5 Group B acceptance. Each sample selected in accordance with 4.4.1.1 shall be examined to determine conformance with the requirements in table V. If one or more defects are found in the inspection sample, the production lot shall be rejected and shall not be supplied to this specification. Group B inspections may be performed in any order acceptable to the government.

4.4.6 Disposition of sample units. Samples subjected to group B tests shall not be delivered on contract or order.

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4.4.6.1 Rejection and retest. When cable selected from a production run fails to meet the specification, no cable still on hand or produced later shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.4.6.1.1 Individual tests may continue. For production reasons, the individual and sampling tests may be continued pending the investigation of a sampling test failure. Final acceptance of cable on hand or produced later shall not be made until it is determined that all cable meets all the requirements of the specification.

4.4.7 Defects in cable already accepted. The investigation of a test failure could indicate that defects may exist in cable already accepted. If so, the contractor shall fully advise the acquiring activity of all defects likely to be found and the method of correcting them.

4.5 Testing and inspections.

4.5.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspections shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with NCSL Z540.3 and ISO 10012-1.

4.6 Methods of inspection.

4.6.1 Test methods. The following identified tests and test methods assure cable integrity within typical operating conditions and application. Alternate commercial industry standard test methods are allowed, however, when an alternate method is used, the qualifying activity must be notified prior to performance of the test. The test methods described herein are proven methods and shall be the referee method in cases of dispute.

4.6.2 Visual inspection. All cable shall be inspected to ascertain compliance with this specification with respect to material, design, and construction, workmanship stranding and marking.

4.6.3 High potential withstanding voltage test. Each reel of cable shall withstand the high potential specified in table I without failure. The test shall be conducted on final shipping reels of cable. For cable supplied on reels containing short lengths of cable, the test may be conducted on reels used for such tests prior to winding on the shipping reel. The high-voltage tests shall be conducted with an alternating current potential from a transformer and generator of ample capacity, but in no case less than 5 kVa. The frequency of the test voltage shall be 60 hertz with a wave shape approximating a sine wave. The initially applied voltage shall be not greater than the rated voltage. The rate of increase shall be approximately uniform and not over 100 percent in 10 seconds nor less than 100 percent in 60 seconds. The reel of cable shall be immersed in a grounded water bath at a temperature of approximately +20 ºC for 6 consecutive hours and, while still immersed, it shall be subjected to the test potential. The potential shall be applied between the conductor and ground. The duration of the tests shall be 5 minutes.

4.6.4 Insulation resistance test. Each reel of cable shall be tested in accordance with NEMA WC 71 to determine compliance with the insulation resistance requirements. This test shall be conducted immediately following the test specified in 4.6.3 .

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4.6.4.1 Insulation thickness test. The average thickness of the insulation shall be determined by measuring the diameter over the insulation and subtracting from it the diameter of the conductor and shield. The difference shall then be divided by two, and the result taken as the thickness of the insulation. Five sets of measurements shall be obtained along the length of a 10-foot sample and the average of the five sets of measurements determined. Each set of measurements shall consist of the determination of the maximum and minimum diameters at the cross-section measured. Measurements shall be made with a micrometer caliper having flat surfaces on both the anvil and end of the spindle, and graduated to read directly in mils, or with a dial micrometer having an anvil, the pressure foot .078 inch wide and .375 inch long, and exerting a force of 10 grams on the pressure foot. If the results of measurement by such methods are in doubt, the optical method shall be used. The optical method utilizes a micrometer microscope accurate to at least .0002 inch (0.01 mm). Measurements of the diameter shall be made at a section halfway between the ends of the sample and at sections 1 inch (25.4 mm) on each side of the midpoint. The average of the maximum and minimum diameters at each section shall be determined, and the lowest of the three averages shall be used as the diameter of the sample in calculating the thickness of the insulation.

4.6.5 Continuity test. Each length of finished cable shall be tested for continuity of the conductor.

4.6.6 Moisture absorption tests. The moisture absorption tests shall be in accordance with part 6 of NEMA WC 71. The test temperature shall be a minimum of 75 ºC.

4.6.6.1 Surface resistivity (dc potential). The samples shall be subjected to the tests specified in NEMA WC 71.

4.6.6.2 Surface resistivity (ac potential). Using the same test specimens, similar electrodes spaced 3 inches (76.2 mm) apart shall be attached over the central test section. A 60-cycle ac potential increasing uniformly to 20,000 volts in 20 seconds shall be applied between the electrodes and maintained for 1 minute without flashover or damage to the cable.

4.6.6.3 U-bend discharge. The samples shall be subjected to the tests specified in NEMA WC 71 except that the test potential shall be 150 volts per mil of nominal thickness.

4.6.7 Ozone resistance test. Samples of finished cable shall show no damage after exposure for 3 hours to an ozone concentration of not less than 0.025 percent and not more than 0.030 percent (by volume) according to the ozone-resistance tests specified in NEMA WC 71.

4.6.8 Cold bend test. A sample of the finished cable shall be placed in a cold temperature chamber and maintained at a temperature of -40 ºC or below for 24 hours. While still in the cold chamber, the sample shall be given five tight turns around a mandrel whose diameter is five times that of the finished cable. There shall be no visible evidence of cracking. The rate of winding shall be approximately five turns per minute.

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4.6.9 Wire resistance test. The resistance of insulated conductor on reels shall be determined at, or the determined resistance corrected to, +20 ºC and shall be expressed as resistance per 1,000 feet. The following formula may be used for converting resistance measured at temperatures other than +20 ºC to resistances at +20 ºC.

$$R_{20} = \frac{254.5R_t}{234.5 + t}$$

Where: R_{20} = Resistance at +20 ºC.

R_t = Resistance at temperature (t), in ohms.

t = temperature at which resistance was measured, in degrees centigrade.

4.6.10 Physical properties test. The physical properties shall be tested in accordance with NEMA WC 71 as follows:

a. Tensile strength (initial and aging).

b. Elongation (initial and aging).

c. Modulus (initial).

4.6.11 High temperature test. The high temperature test shall be conducted in accordance with MIL-STD-810, method 501, procedure I. There shall be no evidence of permanent set or insulation deterioration as a result of this test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification is being retained as a military detail specification because of unique Air Force airport lighting requirements for cross-linked electrical power cable having tin coated conductors. The cable covered by this specification is intended for airport series lighting circuits in direct burial or use on the ground.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class of cable required (see 1.2).

c. If required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

d. Length of cable (see 3.7 and 3.7.1).

e. Inspection conditions, if other than as specified (see 4.2).

f. First article contract requirements.

g. If sample size is to be other than as specified (see 4.3.1 and 4.4.1).

h. Packaging and marking requirements (see 5.1 and 6.3).

6.3 Identification of product. Each reel of cable should be marked for identification. In the past, MIL-STD-130 has been used for marking of U. S. military property. The following information should be applied to each reel of cable:

Voltage rating _____

Size of conductor _____

Number of conductors _____

Length (in feet) _____

Serial number _____ (Reel)

6.4 Subject term (key word) listing.

High voltage
Underground
Environmentally resistant
Weather resistant

6.5 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmental Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website at http://www.epa.gov/osw/hazard/wastemin/priority.htm. Included in the list of 31 priority chemicals are cadmium, lead, and mercury. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein (see section 3).

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6.6 First article. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.7 Tin whisker growth. The use of pure tin may exhibit whisker growth problems (days to months to years) after manufacture. Thin whiskers can develop that uses lead-free pure tin coatings. Conformal coatings applied over top of a whisker-prone surface will not prevent the formulation of tin whiskers. Alloys of 3 percent lead have shown to inhibit the growth of tin whiskers.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
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DLA - CC

Preparing activity:
DLA - CC

Review activity:
Air Force - 99

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